THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE UNIT TEST SET 2

Chapters 5–9: **Exploring Forces, Pressure, Winds, Storms and Cyclones, Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures, The Amazing World of Solutes, Solvents and Solutions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The SI unit of force is the: 1

(a) watt **(b)** joule **(c)** pascal **(d)** newton

2. The SI unit of pressure is the: 1

(a) pascal **(b)** newton **(c)** watt **(d)** joule

3. There are spaces between the particles of matter. Which observation supports this? 1

(a) wood burns
(b) sugar dissolves in water without raising the water level much
(c) ice floats
(d) metals shine

4. Which of the following is an element? 1

(a) H_2O **(b)** $NaCl$ **(c)** CO_2 **(d)** O_2

5. In a solution of sugar in water, sugar is the: 1

(a) solution **(b)** solute **(c)** solvent **(d)** precipitate

6. Which of these reduces friction? 1

(a) lubricants **(b)** dust **(c)** sand **(d)** rough paper

7. Moving air is called: 1

(a) cloud **(b)** wind **(c)** fog **(d)** rain

8. The change from liquid to gas at any temperature below the boiling point is: 1

(a) freezing **(b)** melting **(c)** evaporation **(d)** sublimation

9. The property of metals by which they can be drawn into wires is: 1

(a) solubility **(b)** malleability **(c)** brittleness **(d)** ductility

10. In air, the solvent is: 1

(a) nitrogen **(b)** oxygen **(c)** carbon dioxide **(d)** argon

11. Like charges: 1

(a) attract each other **(b)** do not interact
(c) become neutral **(d)** repel each other

12. The calm centre of a cyclone is called the: 1

(a) arm **(b)** wall **(c)** front **(d)** eye

13. Wet clothes dry faster when: 1

(a) kept in damp room **(b)** tied tightly

(c) spread out in a windy place **(d)** folded and kept in shade

14. In a compound, the elements are combined: 1

(a) in any ratio **(b)** physically only

(c) in a fixed ratio by mass **(d)** without any change of properties

15. 20 g of salt is dissolved in 80 g of water. The mass percentage of salt is: 1

(a) 10 per cent **(b)** 80 per cent **(c)** 25 per cent **(d)** 20 per cent

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why do we sprinkle powder on a carrom board? 2

17. What is atmospheric pressure? 2

18. State two differences between evaporation and boiling. 2

19. What is a metalloid? Give an example. 2

20. State two differences between a solution and a suspension. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Explain the cooling effect of evaporation using particles. 3

22. Explain why water is a compound and not a mixture of hydrogen and oxygen. 3

23. How much solute is needed to prepare 200 g of a 10 per cent salt solution? 3

24. A force of 5 N and another of 5 N act on a book in opposite directions. What is the net force and the effect? 3

OR

Why do we feel the wind more near the sea during the day? 3

25. Why do we see water droplets on the outside of a glass of cold water? 3

OR

What is a molecule? Give two examples of molecules of elements and of compounds. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Describe the characteristics of solutions, suspensions and colloids in a table of five points with examples. 5

OR

Explain the working of a gold-leaf electroscope and what happens when a charged rod touches or is brought near the disc. 5

27. Explain the working of a barometer and the variation of atmospheric pressure with altitude and weather. 5

OR

Describe an experiment to study the cooling by evaporation using cotton wool dipped in different liquids on a thermometer bulb. 5

END OF QUESTION PAPER **Total: 50 Marks**

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